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Temperature Perturbations in the Troposphere and Lower Stratosphere Over a Semi-arid Region During the 2010 Solar Eclipse

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Abstract—Temperature perturbations from the total solar eclipse of January 15, 2010, were monitored from surface to stratosphere (30 km) over Kadapa (14.28° N, 78.42° E), India, a semi-arid region. During the maximum solar eclipse, the entire upper tropospheric and lower stratospheric (UTLS) region reflected the cooling effect. On the eclipse day, the surface temperature showed 2 °C cooling compared to the control day, though surface pressure did not vary. During the eclipse peak, temperature above 30 km altitude decreased by 10 °C compared to the control days. Two inversions in temperature profiles, first around 1 km, near the atmospheric boundary layer, and second near 13 km altitude, were noted after eclipse onset. The diurnal variations in the tropopause temperature, height, and width of the tropical tropopause layer (TTL) were also examined on eclipse and control days. At the time of maximum obscuration on the eclipse day, cold point tropopause (CPT) temperature decreased by $\sim$ 2 °C, and its height decreased by $\sim$ 1.1 km. However, the lower boundary of the TTL did not reflect any significant variations in response to the eclipse. In addition, wave-like oscillations were noted in the UTLS region during the eclipse period. A wave-like pattern with a wavelength of 4–5 km was seen to occur in the stratospheric region (17–22 km) during the maximum eclipse phase. At the end of the eclipse period, the wave pattern speared in the troposphere also (up to $\sim$ 12 km) with a reduced wavelength of 2–3 km.

Keywords: Solar eclipse, temperature perturbations, tropopause dynamics, wave pattern.

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1. Introduction

The solar eclipse is a unique phenomenon that provides an opportunity to investigate the perturbations in the atmosphere associated with rapid changes in solar radiation (Nymphas et al., 2009; Tzanis et al., 2008; and references therein). During a solar eclipse, the Moon's shadow reduces the incoming radiation from the Sun, causing changes in the electron content, neutral composition, and temperature of the Earth's atmosphere. In the past, due to the unavailability of data, most of the studies related to eclipse effects were focused on the lower atmospheric meteorological variables such as air temperature, soil temperature, solar irradiance, humidity, and wind speed (Anderson, 1999; Anderson and Keefer, 1975; Anderson et al., 1972; Antonia et al., 1979; Fernandez et al., 1993, 1996; Hanna, 2000; Zanis et al., 2001; Zerefos et al., 2001). The studies conducted by Antonia et al. (1979) and Szalowski (2002) discussed the changes in the heat and momentum fluxes within the boundary layer during the solar eclipse. Studies by Eaton et al. (1997) and Krishnan et al. (2004) provided evidence for the turbulence processes due to the solar eclipse over Tularosa Basin and Ahmedabad, respectively. The solar eclipse's impact on the atmospheric surface layer characteristics such as air temperature, soil temperature, and wind speed has also been the focus of several studies (Dolas et al., 2002; Founda et al., 2007). For instance, the study by Founda et al. (2007) examined the impact of the total solar eclipse on various meteorological parameters across Greece on March 29, 2006. In particular, their study showed that the decrease in surface air temperature depends not only on the percentage of Sun obscuration but also on several other factors, including the surrounding environment and local cloudiness. Some past studies also concentrated on ozone measurements (Chakrabarty et al., 1997; Gerasopoulos et al., 2007; Prasad et al., 2019; Tzanis et al., 2008; Zerefos et al., 2000). These studies presented a decrease in ozone concentration due to photolysis changes during the eclipse. The gravity wave generation in the lower and middle atmosphere has also been a focus of many previous studies (Abraham et al., 1998; Chimonas & Hines, 1970; Manju et al., 2014; Singh et al., 1989; Voigt et al., 2000; Zerefos et al., 2007).

Most of the abovementioned observational and modeling studies were concentrated on the temperature changes in the tropospheric and stratospheric regions (Ballard et al., 1969; Eckermann et al., 2007; Gerasopoulos et al., 2007), although a few indeed focused on the use of satellite measurements (Kumar et al., 2011; Wang & Liu, 2010). Also, the studies based on direct ground-based observations from the tropospheric region to the stratospheric region are lacking.

The tropopause is a transition layer between the tropospheric and stratospheric regions. The cold point tropopause (CPT) temperature is the coldest point in the vertical temperature profile of the atmosphere below 20 km altitude. The tropopause is one of the critical regions for the upper tropospheric and lower stratospheric (UTLS) dynamics because it controls the exchange between the two regions. Previous studies have linked sudden changes in the CPT to the tropospheric and stratospheric dynamical motions such as the El Niño–Southern Oscillation (ENSO) and Quasi-Biennial Oscillation (QBO) (Hitchman et al., 2021; Kumar et al., 2014a, b; Randel et al., 2003; Schmidt et al., 2005; Zhou et al., 2001). The diurnal variability of the tropical tropopause has also been a focal point of some previous studies (Das et al., 2008). But solar eclipse may be another critical factor for abrupt changes in CPT dynamics for a small time interval. During the annular solar eclipse event on January 15, 2010, a few studies were conducted to understand the eclipse-induced perturbations in the CPT and its height (CPH) (Dutta et al., 2011; Muraleedharan et al., 2011; Subrahmanyam et al., 2011). Muraleedharan et al. (2011) studied the impact of the annular solar eclipse on the tropopause over the coastal region of India (Goa 15° 30′ N, 73° 50′ E). They reported that temperature and CPH decreased during the eclipse day. Subrahmanyam et al. (2011) also confirmed that during the maximum phase of the solar eclipse, the temperature of the CPT decreased and remained at a relatively lower height. After 4 h of the eclipse, the CPT height had drastically reduced by $\sim$ 1.7 km with an increased temperature of $\sim$ 4.6 °C. In a similar study, Dutta et al. (2011) presented the variation of tropopause structure compared to control days; however, this study does not provide any substantial information on tropopause variation during eclipse days. During the solar eclipse on July 22, 2010, using high-vertical-resolution satellite data from COSMIC observations over Kadapa (14.28° N, 78.42° E), Andhra Pradesh, India, a recent study by Prasad et al. (2019) revealed that on the solar eclipse day, the tropopause remained cooler with twin peaks (double tropopause). However, this study mainly focused on variability in atmospheric chemical parameters (O_3 and NO_X) and the temperature structure of the thermosphere between 110 and 130 km.

All the studies mentioned above focused on the variability in the tropopause layer. It is, however, well known that there is no sharp transition from the troposphere to the stratosphere and that the tropopause has a finite width between these two regions. The height of the CPT is known as an upper boundary of the tropopause, while the lower boundary is defined by the top of the convective outflow/convective tropopause (COT). The region between the COT and the CPT is the tropical tropopause layer (TTL). To understand the complete impact of the solar eclipse on tropopause, it is therefore essential to understand the response of the lower boundary of the TTL. The variation in TTL width in response to solar eclipse has remained an unexplored topic during previous studies. This paper presents comprehensive and substantial information on the temperature of CPT, CPH, and TTL width during the solar eclipse of January 15, 2010. This study highlights the findings on temperature perturbations and possible generations of the waves in tropospheric and stratospheric regions, particularly in a semi-arid region.

Semi-arid regions have very low precipitation, high evaporation, and low vegetation, and the surface of these regions is generally dry and mainly covered by sand and dust. This makes the semi-arid region different, as the heat capacity of the dust surface is low, and hence the temperature will be more sensitive to incoming radiation. During the solar eclipse period, it is expected that the temperature over a semi-arid region will reflect more sensitivity than a non-semi-arid region. In addition, vertical atmospheric variability over semi-arid regions also shows a fast response to surface temperature variability. In this context, it is pertinent to mention that semi-arid regions have not been studied for the eclipse impact on temperature variability in the tropospheric and stratospheric regions. Hence, the present study focuses on the semi-arid region Kadapa (14.28° N, 78.42° E) in the southern part of India for studying variations in the surface, tropospheric, and stratospheric temperatures. Though our observational site Kadapa does not come directly under the path of eclipse totality, it is located only slightly away from the path such that the eclipse impact could be easily accessed. In the following, we discuss the results in detail.

2. *Data and Methodology*

To investigate the impact of the annular solar eclipse of January 15, 2010, on meteorological parameters, especially on the temperature of the UTLS region, a particular field campaign was conducted over Kadapa (14.28° N, 78.42° E), India, from January 12 to 15, 2010. We used Pisharoty radiosondes for this purpose in the campaign. These are GPS-based reliable and efficient systems, well suited for atmospheric studies, weather prediction, and other related aerospace applications (Choudhary et al., 2013; Divya et al., 2014). Pisharoty radiosondes were launched to measure the temperature in the tropospheric to stratospheric region. The first two versions of the Pisharoty sondes showed significant radiative heating because of the platinum resistance temperature detector used in the temperature sensor. The third (latest) operational version (PS01_B3) of the sonde was developed with a very small glass bead thermistor as a temperature sensor, with solar and infrared radiation corrections in ground software (Divya et al., 2014). The PS01_B3 version of Pisharoty GPS radiosondes has been used in this study to avoid any solar heating-related issues. It is worth mentioning that the temperature and pressure measurements from these sondes match fairly well with the data from other established sondes (e.g., Vaisala, Meisei, and Graw) with a high correlation coefficient of 0.999 (Choudhary et al., 2013; Divya et al., 2014). Details of the specification, observations of meteorological parameters, and accuracy of Pisharoty sondes are described by Divya et al. (2014).

Pisharoty GPS radiosondes at Kadapa were launched every 3 h for 3 days from January 12 to 15, 2010. Data from ten meteorological balloon flights, one on January 12, 2010 (11:30 IST), four on January 14, 2010 (08:30, 14:30, 17:30, and 20:30 IST), and five on January 15, 2010 (08:30, 11:30, 14:30, 17:30, and 20:30 IST) are utilized to understand the temperature perturbations of the UTLS region in response to the solar eclipse. Each flight took $\sim$ 2.00 h to travel from the surface to the stratospheric region at $\sim$ 30 km. Since the balloons launched at 11:30 IST on January 13 and 14, 2010, could reach a maximum height of only 14 km, we have considered data from January 12, 2010, as a control day profile for 11:30 IST.

On January 15, 2010, we released the first (at 08:30 IST) and second (at 11:30 IST) balloon flights in such a way that the radiosonde should have been in the upper atmosphere over or nearby Kadapa during the beginning (11:23 IST) and peak phase of the solar eclipse (13:28 IST), respectively. To study atmospheric behaviors during the entire eclipse day, balloons were also launched at 14:30 IST (ending phase of the solar eclipse), 17:30 IST, and 20:30 IST. During each flight, all the essential meteorological parameters such as temperature, pressure, humidity, and winds were measured from surface to 30 km height with an altitude resolution of $\sim$ 10 m (sampled at 2-s intervals). Later, the entire data set was suitably averaged to 100 m.

Automatic weather station (AWS) facilities are available on the campus of Yogi Vemana University, Kadapa (14.28° N, 78.42° E), India. AWS measurements are accessible for surface meteorological parameters, such as air temperature, pressure, and rainfall, every 4 min. Hence, the AWS data were also used for surface air temperature and pressure from January 14 to 16, 2010, to capture the effect of the eclipse in the preceding and following days. The causative factors for such variability could be investigated. All these observational results are presented in the next section.

3. Results and Discussion

3.1. Influence of Solar Eclipse on Temperature Structure from the Surface to Stratospheric Region

Solar eclipses provide a rare opportunity to study the response of the Earth's atmosphere to a step function kind of input. Depending upon its time of occurrence, duration, and geographical location, each solar eclipse uniquely influences the Earth's atmosphere. The main objective of this study is to understand the effect of the longest annular solar eclipse, a rare and unique phenomenon on January 15, 2010, on the energy of the TTL region. The impacts of this solar eclipse on temperature structure are examined from the lower to the upper atmosphere. The observational site, Kadapa (14.28° N, 78.42° E), is a semi-arid station in India situated near (< 300 km) the path of totality, and the location of the station has some unique features in terms of atmospheric characteristics, as described briefly in the introduction.

The total solar eclipse of January 15, 2010, over Kadapa started at 11:23 IST (IST $\sim$ UTC + 05:30) with maximum obscuration ($\sim$ 76%) at 13:28 IST and ended at 15:14 IST. The total duration of the event was 3 h and 50 min, with an obscuration of 76% at the time of the maximum phase ($\sim$ 13:28 IST). Obscuration of the solar eclipse is the fraction of the Sun's area occulted by the Moon. Due to 76% obscuration, Kadapa can be considered a suitable location for understanding solar eclipse-induced perturbation in the Earth's atmosphere bereft of complete obscuration. The solar eclipse path on January 15, 2010, over the Indian region is shown in Fig. 1. This figure is based on the data provided by Fred Espenak, NASA's Goddard Space Flight (GSF).

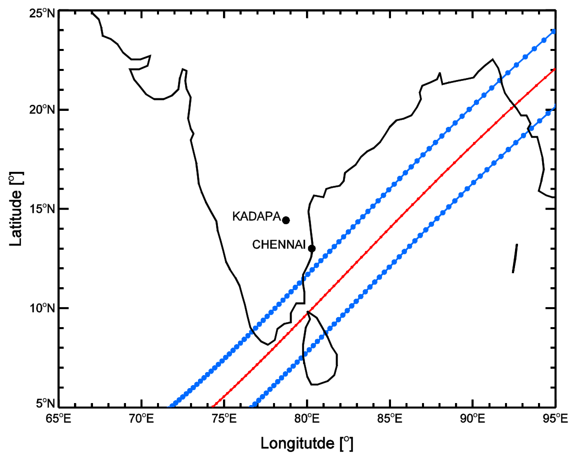


Figure 1
Path of the solar eclipse during January 15, 2010, over the Indian region. Blue and red lines represent the edges and center of the Moon's umbra shadow, respectively, during the eclipse period

Blue and red lines represent the edges and center of the Moon's umbra shadow during the solar eclipse period, respectively. This eclipse was exceptional in having a track crossing densely populated land masses and occurring near mid-day. It thus provided a rare opportunity to assess the influence of the solar terminator's supersonically moving waxing and waning on meteorological parameters, like wind and air temperature.

Using AWS data from January 14 and 15, 2010, the hourly averaged diurnal variation of surface air temperature and pressure are shown in Fig. 2. Three vertical lines, marked at 11:23 IST, 13:28 IST, and 15:14 IST, respectively indicate the beginning, peak, and ending timings of the solar eclipse. Figure 2a illustrates the hourly averaged diurnal variation of surface air temperature on January 14, 15, and 16, 2010, in blue, red, and green lines, respectively. The temperature exhibited a typical daily pattern on January 14 and 16, 2010, with maximum temperatures of 32 °C and 30 °C, respectively, at 15:30 IST (Fig. 2a). However, as the eclipse set in on January 15, 2010, an apparent decrease in the surface air temperature can be seen compared to an expected behavior with its minimum magnitude at the time of maximum obscuration during the peak period of the eclipse.

Interestingly, the decrease in temperature continued even after the eclipse period. The drop in the

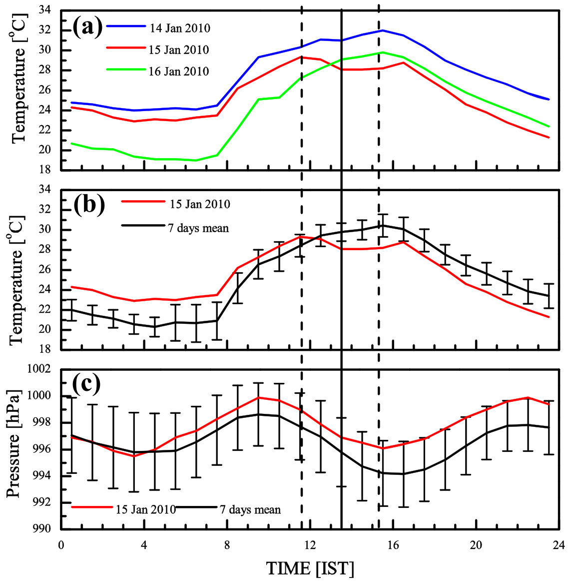


Figure 2
Diurnal variation of surface air temperature measured by AWS **a** from January 14 to 16, 2010, **b** on eclipse day (red line), and 7-day mean with standard deviation (black line). **c** Same as in part **b** but for surface pressure. Three vertical dashed lines marked at 11:23 IST, 13:28 IST, and 15:14 IST indicate the timings of the beginning, peak, and the ending of the total solar eclipse, respectively

surface air temperature was ~ 4 to 6 °C when compared to the preceding (January 14) and the following (January 16) day. To quantify the apparent effect of the solar eclipse, the diurnal variation of surface air temperature on eclipse day and monthly mean temperature is shown in part (b) of Fig. 2 (middle panel). In January 2010, we had 7 days of observations to estimate the control day mean. In Fig. 2b, the red line represents the diurnal variation of surface air temperature on January 15. The black line represents the 7-day mean and the standard deviation. It is amply clear from Fig. 2b that on January 15, 2010, after the beginning of the eclipse, the surface air temperature was cooler by ~ 2 °C in comparison to the 7-day mean variation. These observational results indicate solar eclipse-induced cooling perturbations in the surface air temperature. The surface pressure variability was also investigated and is shown in Fig. 2c (bottom panel). Again, the red line represents the diurnal surface pressure variation during January 15, and the black line is the 7-day mean and the standard deviation. On the eclipse day, after 05:00 IST, though the surface pressure was slightly higher than the 7-day mean, this higher value lies within the standard deviation range.

We extended our study using GPS radiosonde measurements to investigate the temperature-induced eclipse-induced perturbations from the surface to 30 km. The analyses are shown in Fig. 3. Part (a) of Fig. 3 shows the vertical temperature profiles during January 14 and 15, 2010, at 08:30, 11:30, 14:30, 17:30, and 20:30 IST in different line styles. The two dates are shown in blue and red colors, respectively. Successive vertical profiles have been displaced by 10 °C for clarity. In the troposphere, the temperature is almost stable, but in the stratospheric region, instability can be seen in the temperature profile (Fig. 3a).

Interestingly, two inversions appeared in the temperature profiles after the onset of the solar eclipse, the first around 1 km, near the atmospheric boundary layer, and the second near 13 km altitude. A previous study by Muraleedharan et al. (2011) showed similar inversions in Goa, India. The present study confirms their findings. The occurrence of temperature inversions indicates more substantial stability which inhibits vertical movement of air just above the atmospheric boundary layer and in the upper troposphere (Muraleedharan et al., 2011). It is also noteworthy that in the stratospheric region above 20 km, despite 10 °C successive displacements of profiles, the 17:30 IST temperature profile overlaps with the 14:30 IST profile on both days, i.e., January 14 and 15, 2010. This may be attributed to the night effect (evening transition) in Northern Hemispheric winter. As each flight took ~ 1 h to reach the stratospheric region (~ 20 km), the 17.30 IST flight was expected to enter the stratospheric region during the night. The absence of sunlight leads to a reduction in ozone heating, and the temperature of the stratospheric region decreases sharply.

Another interesting point to note is the fact that the 11:30 IST temperature profile also overlaps with the 08:30 IST profile on the eclipse day. As discussed above, that 11.30 IST flight will be in the upper atmosphere during the peak time of the solar eclipse, and one can quickly notice the cooling by ~ 10 °C in the stratosphere. Two factors are mainly responsible for such drastic cooling in the stratospheric

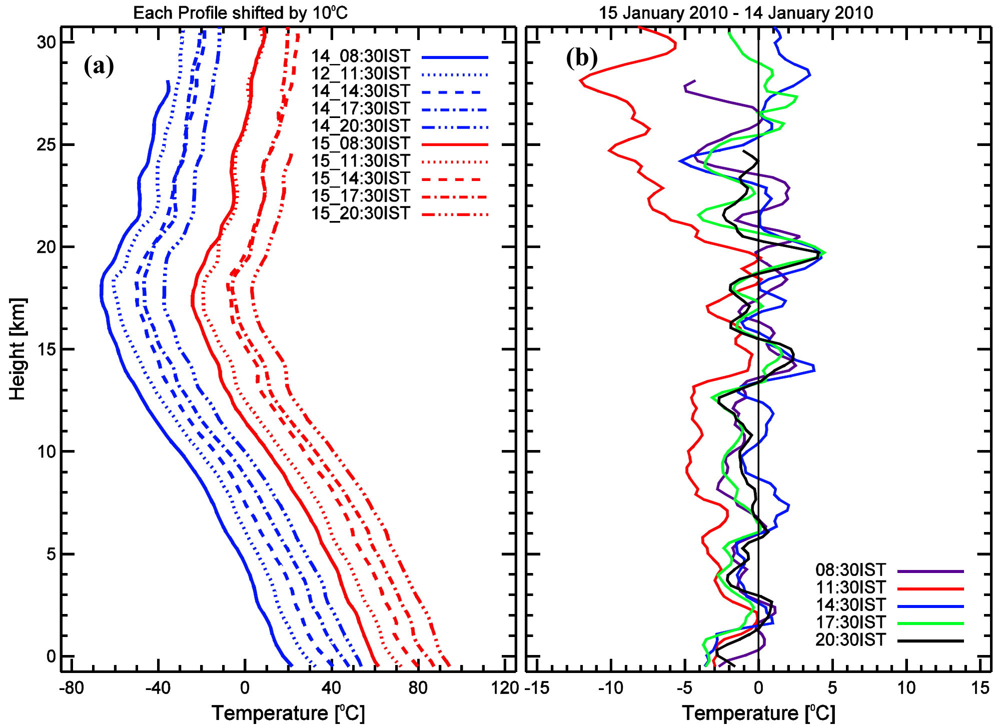


Figure 3

a Altitude profile of temperature up to 30 km for January 14, 2010 (blue color), and January 15, 2010 (red color), during the different radiosonde flights; i.e., 08:30 IST, 11:30 IST, 14:30 IST, 17:30 IST, and 20:30 IST. Each profile is shifted by 10 °C for clarity. **b** Altitude profiles of temperature perturbation obtained by subtraction of the eclipse day profile from the corresponding flight of the same time on a control day. Since the balloon launched at 11:30 IST on January 13 and 14, 2010, reached the maximum height of 14 km, January 12, 2010, is considered a control day profile for 11:30 IST only

region. First, the ozone heating declines during the solar eclipse period due to occultation in solar radiation, and second, the mixing ratio of O_3 also decreases due to photolytic changes during the eclipse period (Prasad et al., 2019).

As each successive profile in Fig. 3a is displaced by 10 °C to capture small temperature changes, the profiles of eclipse day were subtracted from the corresponding profiles of the control day, and these deviations are shown in Fig. 3b. Positive and negative values indicate warming and cooling, respectively, in temperature during the eclipse day compared to the control day. In the 11:30 IST profile, a slight decrease in temperature can be observed in the entire lower atmosphere from the surface to 30 km, as the flying time of this flight lies within the vicinity of the eclipse period.

To understand the strength of perturbation in temperature during the eclipse period, the average of three daytime profiles at 08:30, 11:30, and 14:30 IST on the control day and the 08:30 IST profile of the eclipse day is shown with a black line in Fig. 4a. We have not included the 17:30 IST and 20:30 IST profiles of both days to avoid the night. Along with this mean profile, the profiles of 11:30 IST (red line) and 14:30 IST (blue line) on eclipse day are also shown. We may note that between the altitude region of 15 and 30 km, the temperature profile of 11:30 IST differs significantly from the mean profile. Towards the end of the eclipse (15:14 IST), as the 14:30 IST flight reached the stratosphere, the temperature profile regained its controlled day values and followed the mean temperature structure with small perturbations at different levels. We could not detect high temperatures over the stratospheric region, as Wang and Liu (2010) reported.

Further, to see more fine-scale perturbations in temperature profiles, we subtracted the mean profile

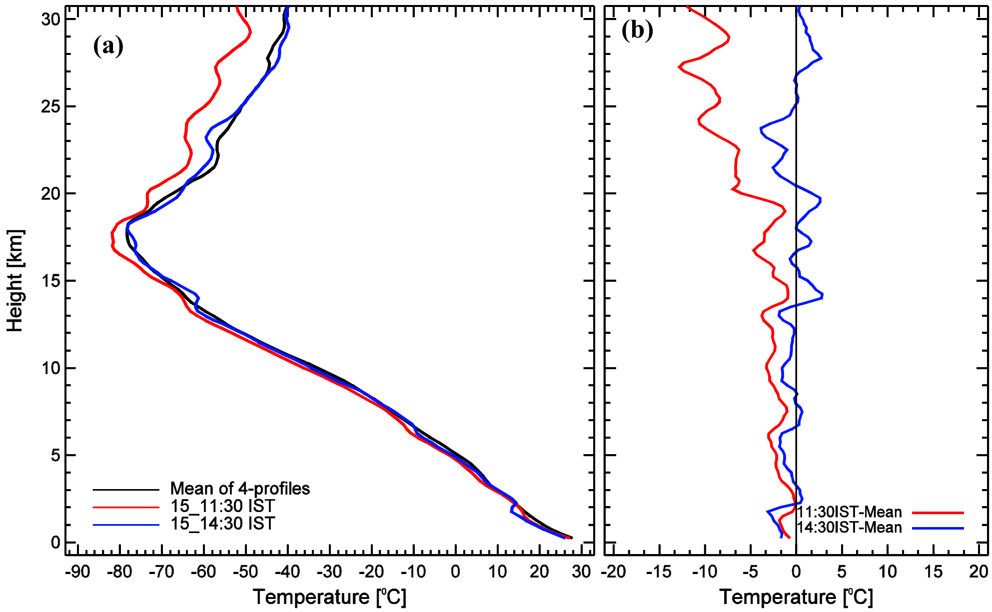


Figure 4
a Altitude mean temperature profile of four sonde flights (i.e., 08:30, 11:30, 14:30 IST of January 14 and 08:30 IST of January 15) along with temperature profiles observed during the eclipse (i.e., 11:30 and 14:30 IST). **b** Altitude profiles of temperature perturbation obtained by the subtraction of mean profile from 11:30 IST (red) and 14:30 IST (blue)

from 11:30 IST (red line) and 14:30 IST (blue line) profiles, and the mean subtracted profiles are shown in Fig. 4b. A clear wave-like increasing cooling pattern from the surface to 30 km can be observed in the 11:30 IST profile. The dominance of these cooling patterns increases from the troposphere (~ 2–4 °C) to the stratosphere (~ 10–12 °C). The 14:30 IST profile does not show much variation and follows the zero-degree line.

To understand systematic changes in stratospheric temperature as the eclipse evolves, each profile on eclipse day, subtracted from the preceding profile in time, is shown in Fig. 5. When the 08:30 IST profile is subtracted from the 11:30 IST profile (red line), it shows a strong negative deviation (up to ~ −10 °C) as the entire flight of 11:30 IST lies within the vicinity of the eclipse period. On the other hand, the subtraction of 11:30 IST from the 14:30 IST profile (blue line) exhibits a substantial positive deviation (~ 10 °C). The subtraction of the 14:30 IST profile from 17:30 IST (green line) shows a negative deviation (−15 °C) due to the night effect, as discussed above. Lastly, the black line, which shows the subtraction of the 17:30 IST temperature profile from 20:30 IST, reflects almost zero deviation. This zero deviation reveals that no more temperature variations were observed, as the eclipse period was over, like any other ordinary/control day.

We note that during the period of eclipse (11:30 IST to 2:30 IST), there were significant temperature variations in the stratospheric region. Strong temperature perturbations in the stratospheric region may disturb existing stratospheric meridional circulation. The new circulation may influence the dynamics of the stratosphere over stations located far from the path. Dhaka et al. (2015) have discussed such a type of stronger meridional stratospheric circulation during the sudden stratospheric warming (SSW). Their study found that polar and tropical regions are highly coupled during the SSW, most likely through the rapid setup of strong stratospheric meridional circulation within 1 day. Such aspects need to be explored during the solar eclipse period as the stratospheric region shows substantial temperature variability. We shall take up this as our future study using global data from satellite measurements.

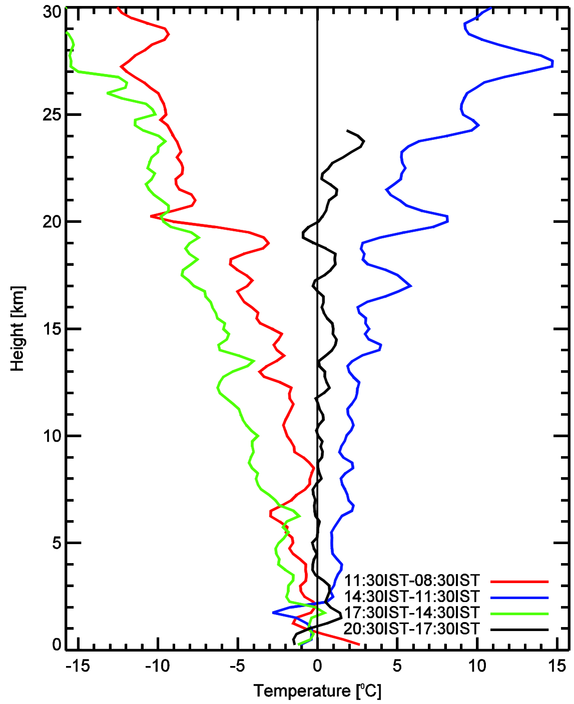


Figure 5
Systematics perturbations in altitude profile of temperature during eclipse day. These profiles were obtained by subtracting each profile from the preceding profile

3.2. *Impact on Tropopause Dynamics*

As discussed above, the tropopause is one of the critical regions for UTLS dynamics. To explain variations in the tropopause temperature during the solar eclipse, the diurnal variations of cold point tropopause (CPT), cold point tropopause height (CPH), and tropical tropopause layer (TTL) on the eclipse and control days are shown in Fig. 6. The top panel (a) of Fig. 6 represents the CPH and lower boundary of the TTL. The top boundary of the TTL is the CPH itself. Hence, the CPH and TTL width variation can be observed simultaneously in Fig. 6a. The bottom part (b) of Fig. 6 denotes the CPT. At the beginning of the solar eclipse, the CPT dropped by ∼ 2 °C and was confined at a relatively lower height (∼ 1.1 km) than the control day. After the solar eclipse, the patterns of CPT with a slight dip value (∼ 1 °C) show the same variability as on the control days but at a relatively higher height (∼ 0.8 km). However, we have not observed any significant changes in the lower boundary of the TTL; only a slight decrease (∼ 100 m) can be viewed during maximum obscuration, which otherwise remained constant. The lower boundary of the TTL almost coincides and follows the same pattern on eclipse day and control day.

In a related study, Subrahmanyam et al. (2011) showed a drastic decrease in the CPH and an increased CPT value during the post-eclipse period. They obtained this decrease in the CPH without much change in the troposphere and related this finding to the adiabatic compression of the lower stratospheric atmosphere due to the solar eclipse. However, our study does not support this mechanism because no such drastic changes were observed at our location. We surmise that there could be another local factor for drastic changes in their study.

Interestingly, in our observations, the lower boundary of the TTL does not show any significant variations during the eclipse period. This indicates that stratospheric variability may not be the source influencing the CPH (TTL upper boundary); otherwise, the lower boundary of the TTL should also have shown significant perturbations as the tropopause is directly connected with the lower stratospheric region. As discussed, vertical atmospheric variability over semi-arid regions shows a fast response. During the peak period of the eclipse, due to a drop in surface air temperature, the convection should be slightly weaker than in the non-eclipse period. Hence, upward forcing will be somewhat weak, allowing settling of the tropopause at a lower height. However, the minimum potential temperature gradient indicates the lower boundary of the TTL (Gettelman and Forster, 2002). The potential temperature is less sensitive to any small-scale changes in the convection. This could be why the lower boundary of the TTL does not show any significant variation. This study does not support the mechanics suggesting that during the eclipse day, lowering of the tropopause height is associated with downward forcing of the stratospheric air mass to the denser troposphere, and this process would warm the tropopause and the upper tropospheric region, as discussed by Muraleedharan et al. (2011). This is for the simple reason that we did not observe a warmer CPT at our station.

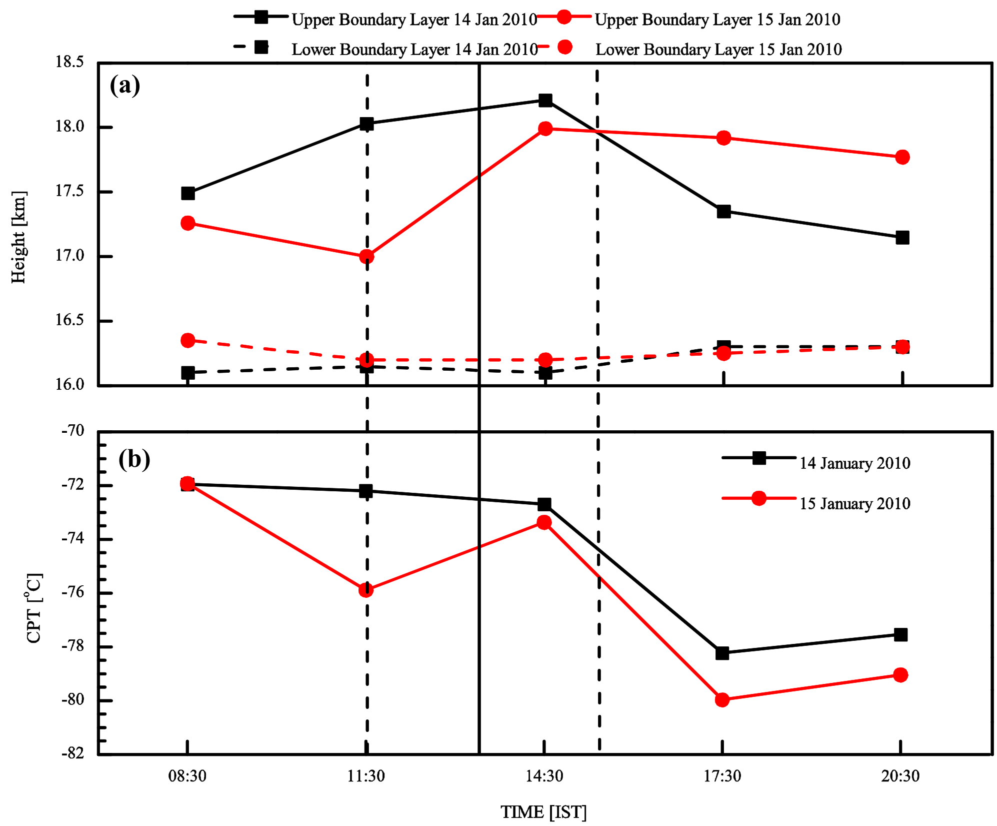
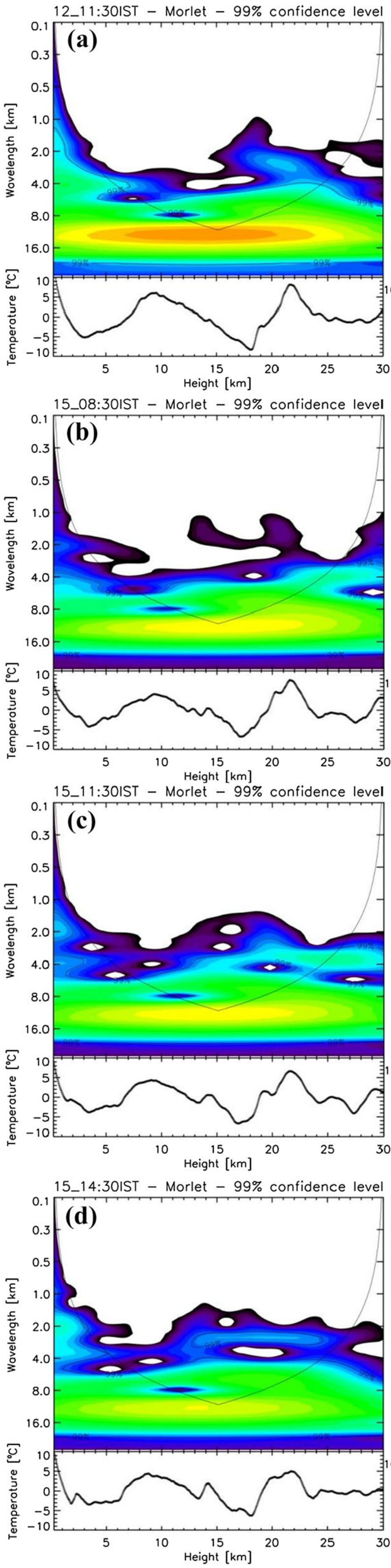


Figure 6
Diurnal variation of **a** the CPH and TTL and **b** CPT during January 14 and 15, 2010

3.3. *Wave Activity*

Wave activity is an essential part of the dynamics of the Earth's atmosphere. An attempt has been made to investigate the solar eclipse's influence on these wave activities using the continuous wavelet transform (CWT) which divides a continuous time function into wavelets. Unlike Fourier transform, the CWT can construct a time–frequency representation of a signal that offers a perfect time and frequency localization. Farge (1992) and Torrence and Compo (1998) provided a complete description of the CWT. The program is written in interactive data language (IDL), provided by Torrence and Compo, and is used for the present analysis. The Morlet wavelet ($w_0 = 6$) is used as a mother wavelet, a plane wave modulated by a Gaussian feature. The Fourier spectra (FS) were also computed for comparison.

The CWT becomes a more powerful tool to simultaneously understand geophysical phenomena in terms of frequency and time domains. The CWT was applied to the temperature profiles of 08:30 IST, 11:30 IST, and 14:30 IST on the control and eclipse days. The wavelet plots show the same feature on the control day for each time, i.e., 08:30 IST, 11:30 IST, and 14:30 IST. Here, we show only the 11:30 IST wavelet plot as an example for the control day. We have not included the nighttime profiles because no variation was observed at night compared to the control and eclipse day. But some variations were observed at different periods during the eclipse day. The scalograms for 11:30 IST of the control day and 08:30 IST, 11:30 IST, and 14:30 IST of the eclipse day are shown in Fig. 7a–d. Each part of Fig. 7 contains three panels. Before applying the CWT, each profile's fitted linear regression line was



◀Figure 7

a–d The wavelet plots of **a** 11:30 IST on control day, and **b** 08:30 IST, **c** 11:30 IST, and **d** 14:30 IST on eclipse day. The thick solid U-shape line in each wavelet represents the COI). The *x*-axis of each wavelet represents the height, and the *y*-axis wavelength. The confidence level of each wavelet spectra is of the order of 99%. Series at the bottom of each wavelet represents the temperature after removing unwanted data (with one sigma variation) and was used to compute the wavelet power spectrum. The plot on the right contains the global power spectrum of the entire profile from 0 to 30 km, with the 99% confidence level shown as a continuous red line. Stars also show the Fourier spectrum (FS) in a black line for comparison. The color code of the wavelet is shown at the top left side of each panel

separated from its corresponding profile, which removed the unwanted components available in each profile.

The resultant profile resembled a standard deviation of that profile. The temperature, after removing unwanted data, is shown at the bottom of the left panel in Fig. 7a–d. This temperature was used to compute the wavelength spectra, shown at the top of the left panel (Fig. 7a–d) with the color code. The *x*-axis of wavelength spectra represents the height (km), and *y*-axis wavelength. The thick solid U-shape line in wavelength spectra represents the scalogram's cone of influence (COI). The COI is considered a parameter to describe the wave because wavelet coefficients outside this COI are unreliable due to edge effects. The confidence level of each power spectra is at 99%. The vertical plot in the red line on the right of Fig. 7a–d represents the global power spectrum obtained by averaging the wavelet power over the entire height level, i.e., from 0 to 30 km. Again, the results are at a 99% confidence level for these GPS spectra. The Fourier spectrum (FS) is also shown in the black star line; this is only for comparison. The detailed comparison between the wavelet and the FS can be found in Torrence and Compo (1998). It is seen from Fig. 7a that there is no oscillatory wave feature in the 11:30 IST profile of the control day, while small oscillations can be seen in the 08:30 IST profile of the eclipse day (Fig. 7b). During the maximum phase of the solar eclipse profile (i.e., 11:30 IST), significant wave oscillations are observed in the lower stratosphere around ~ 17–22 km height with ~ 5 km vertical wavelength (Fig. 7c). Further, these wave oscillations extend in the upper troposphere and lower stratosphere (13 to 24 km), but the wavelength reduces to 3 km (14:30 IST, Fig. 7d).

CWT analyses suggest that wave-like structures emerge in the temperature perturbation during the maximum eclipse phase in the lower stratosphere. These wave-like structures are attributed to the gravity wave induced by the solar eclipse.

4. Summary and Conclusions

Integrated atmospheric measurements were made at Kadapa (14.28° N, 78.42° E), India, during the annular solar eclipse of January 15, 2010. Short-term eclipse-related changes associated with the synoptic conditions dominated over temperature from the surface to the stratosphere. After the beginning of the eclipse, the air surface temperature was low throughout the day compared to the control days. However, surface pressure data did not show any clear impression regarding the solar eclipse. As the solar eclipse started, small inversions appeared in the temperature structure at 1 km and 13 km altitude. A similar feature was obtained by Muraleedharan et al. (2011) over a different observation site in Goa, India. The stratospheric region reflected more substantial temperature variation during the eclipse day than on preceding and following days. Such a significant variability of the stratospheric region may disturb existing stratospheric meridional circulation or may also generate new modified circulation; this deserves more detailed study in the future. Current observations support the findings of previous studies (Muraleedharan et al., 2011; Subrahmanyam et al., 2011) that during the maximum phase of the solar eclipse, the temperature of CPT decreases and occurs at a relatively lower height. However, the present study could not observe any delayed effect of the eclipse on the tropopause as reported in a previous study (Muraleedharan et al., 2011). The current study has shown some new features related to the impact of the solar eclipse on the TTL, which had remained unexplored during previous studies. No significant changes in the lower boundary of the TTL were observed in response to the eclipse. Using the CWT, we also attempted to find out the influence of solar eclipse on the wave activities in the UTLS region and during the maximum phase of the solar eclipse. A wave pattern appeared in the height range of 17–22 km which deserves a more detailed future study.

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Declarations

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